

Planetary Radius as a Regime-Dependent Relational Quantity: Evidence for a Paradigm Defect in Exoplanet Radius Measurement

——A Cross-Disciplinary Empirical Test in Astronomy Based on Factor Hierarchy Theory and the Four-Test Method

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Abstract

Core Conclusions: (1) Transit depth is not an intrinsic property of a planet, but a “relational quantity” arising from the coupling of the planet, the stellar radiation field, and the detector response. (2) The discovery facility is a proxy regime factor for the stellar radiation field—the true regime factors are stellar spectral type, metallicity, and surface gravity. (3) The concept of “planetary radius” as used in astronomy for four decades lacks physical universality in cross-regime comparisons. This study applies the Principle of Testability and the Four-Test Method proposed by Tang (2026g), augmented by the 5W2H and PDCA completeness frameworks, to a presupposition-free exhaustive analysis of 39,913 raw records from the NASA Exoplanet Archive. Exhaustive testing automatically selected `radius_ratio + pl_orbper + sy_dist` as the optimal combination from 12 candidate executing factors (adjusted $R^2 = 0.691$, $VIF < 5$). Interaction effect testing revealed that the discovery facility systematically modulates the exposure coefficient of `radius_ratio` on transit depth ($\Delta R^2 = 0.048$, $p = 6.59 \times 10^{-7}$); however, this effect completely vanishes when controlling for stellar spectral type—G-type: $F = 0.05$, $p = 0.99$; K-type: $F = 1.01$, $p = 0.39$. This finding reveals a new structural form within Factor Hierarchy Theory: the proxy regime factor. Stellar spectral type, metallicity, and surface gravity were all tested and confirmed as significant regime factors. The regime coupling is strongest in the metal-poor group ($\Delta R^2 = 0.0313$, $p = 1.21 \times 10^{-15}$) and disappears in the metal-rich group ($p = 0.428$). A temperature threshold of 4762 K ($F = 10,283$) and a surface gravity threshold of $\log g = 4.64$ ($F = 14,645$) constitute a double critical point for regime switching. The number of planets observed by both Kepler and TESS is zero. Permutation tests (empirical $p = 0.002$), quantile regressions (all quantiles $p = 0$), and pseudo-out-of-sample prediction (RMSE amplification of $4.4\times$) comprehensively support the conclusions. This paper proposes a procedural criterion: before merging multi-facility transit data, one should conduct interaction and Chow tests using candidate regime factors as grouping variables; if both are significant, the regime switch should be acknowledged, and separate modeling should be adopted rather than forced uniform calibration. Given that Factor Hierarchy Theory has

independently replicated the same hierarchical structure across five financial markets and exoplanetary transit measurements, this paper proposes elevating this theory to the status of Factor Hierarchy Law. The empirical process of this paper also completes the first cross-disciplinary implementation of the 5W2H and PDCA frameworks as completeness companions to the Four-Test Method.

Research Paradigm Statement: The core methodology, research direction, and final decisions were independently directed by the author. DeepSeek assisted with code implementation, data presentation, and text drafting. The author takes full academic responsibility for the final content.

Keywords: Transit-Depth Precision Problem; regime coupling; proxy regime factor; Factor Hierarchy Law; Four-Test Method; testability; planetary radius

1. Introduction

1.1 The Transit-Depth Precision Problem: A Gap That Grows With Every Fix

Exoplanet science faces a disturbing paradox. As space telescopes such as Kepler, K2, and TESS have accumulated vast quantities of transit data, astronomers have found that the measurement uncertainty of transit depth has not converged but instead systematically deviates from photon-noise expectations. The recently articulated “Transit-Depth Precision Problem” (TDPP) explicitly states that actual errors can be amplified to over ten times the theoretical value, and this amplification cannot be explained by existing noise models.

Wang & Espinoza (2022) analyzed 28 known planets and showed that transit depths measured by different facilities or observers exhibit significant differences. Morello et al. (2020), in multi-wavelength transit observations using Spitzer and HST, found significant transit depth differences between instruments, with the prevailing interpretation attributing these to stellar activity and inadequacies in limb-darkening models. The astronomical community's response has been to adopt more complex limb-darkening (LD) models, add fitting parameters, and employ higher-order polynomials. However, the core logic of these approaches remains one of “correction”—treating the differences between facilities as **systematic errors** that need to be calibrated away.

1.2 A Question That Has Never Been Asked

This paper proposes an entirely different diagnosis. We do not ask “how to better correct for facility differences.” Instead, we ask a more fundamental question: **Should facility differences be corrected at all?**

If the differences between facilities are not linear, calibratable errors but rather structural, irreducible regime switches, then any effort at “correction” rests on a false premise. Astronomers may have pushed further down the road of “model fixing” precisely because they have never questioned the premise itself: that **“planetary radius” is an objective property that exists independently of the measurement regime.**

The task of this paper is to subject this never-before-questioned premise to systematic empirical testing using a methodology that is entirely independent of astronomy. The Four-Test Method originates in finance and shares none of astronomy's presuppositions—this “external test” possesses a power that methods internal to astronomy cannot replicate. This independence is not merely rhetorical: a methodology that does not share a discipline's unexamined defaults is the only kind capable of detecting those defaults.

1.3 Theoretical Framework: Factor Hierarchy Theory and the Principle of Testability

The methodological foundation of this study comes from the **Principle of Testability** and **Factor Hierarchy Theory** proposed by Tang (2026g). The theory posits that in any complex system driven by multiple quantifiable factors, factors naturally fall into two hierarchical levels: **regime factors** determine which set of rules currently applies, and **executing factors** operate under the given rules. The function of a regime factor is not reflected in its direct explanatory power, but rather in the interaction effects through which it modulates the exposure coefficients of executing factors. When a regime factor crosses a critical threshold, the factor weights of the entire system are systematically reset.

Tang (2026g) discovered in finance that the China-U.S. interest rate spread is not a traditional pricing factor (its direct R^2 is near zero) but rather a **regime factor** that determines how executing factors such as VIX and DXY explain asset prices. When the spread crosses the zero axis, the entire pricing equation undergoes a structural break. This theory has been fully tested across five major financial markets, but independent cross-disciplinary verification remains pending.

At the level of philosophy of science, Tang (2026g) points out that Popper's falsifiability stipulates the **entry criterion** for scientific knowledge—a proposition must be capable of being overturned by facts. However, the classic criticism of Popper's theory within philosophy of science has always focused on its “attitude principle without operational procedures”: Popper required scientists to possess the spirit of falsification, but he never specified “what counts as rigorous testing” in concrete empirical research. Tang

(2026g) proposes that science requires a **second threshold—testability**: a procedural standard that specifies the concrete steps of testing—exhaustive search replaces prior selection, interaction effects and Chow tests replace the assumption of coefficient stability, and multiple verification replaces single significance. Falsifiability guards the entrance of science, testability guards the exit of scientific discovery. Together, they constitute the complete chain of scientific methodology.

1.4 5W2H and PDCA: Completeness Companions to the Testing Procedures

The Four-Test Method of Tang (2026g) specifies the **procedures** of testing—exhaustive search, interaction effects, Chow tests, and auxiliary evidence. But procedures alone do not guarantee completeness—how does one judge whether a research question has been exhausted in all its dimensions? How does one judge whether the research process forms a closed loop? Tang (2026g) reserved an interface at the methodological level for “completeness companions to the testing procedures.”

This paper is the first to introduce the **5W2H** and **PDCA** frameworks from management science as the concrete implementation of this interface. 5W2H (What, Why, Who, Where, When, How, How Much) ensures that the research question is completely covered across seven dimensions; PDCA (Plan, Do, Check, Act) ensures that the research process completes the full Plan-Do-Check-Act cycle. The three together constitute the complete operational system of the Principle of Testability: the Four-Test Method provides the testing procedures, 5W2H provides the standard of question completeness, and PDCA provides the standard of process closure. This paper is the first cross-disciplinary full-implementation case of this companion system.

2. Data and Methodology

2.1 Data Source

All data in this study come from the NASA Exoplanet Archive (<https://exoplanetarchive.ipac.caltech.edu>), downloaded via the TAP synchronous query interface:

```
curl -o "confirmed_planets.csv" "https://exoplanetarchive.ipac.caltech.edu/TAP/sync?query=select+*+from+ps&format=csv"
```

The raw file contains **39,913 rows and 355 columns**, constituting the full table of confirmed and candidate exoplanets maintained by NASA. This study performed **no pre-screening of the raw file**—all screening was carried out transparently within Python code, and the raw data remains intact. Any researcher can download the same file and independently reproduce all results using the methods described herein.

2.2 Data Cleaning

Sample selection: Only planets discovered by the transit method were retained (**35,776**). Transit depth (`pl_trandep`) must exist and fall within the physically reasonable range (>0 and $<100\%$). The planet-to-star radius ratio (`radius_ratio`) must be less than 0.5 to eliminate contamination from stellar companions or brown dwarfs.

Radius ratio correction: During data auditing, we discovered a unit inconsistency between `pl_rade` (planet radius, in Earth radii) and `st_rad` (stellar radius, in Solar radii), causing the direct ratio to yield physically impossible values (median as high as 2.56). The correction formula adopted is: $\text{radius_ratio} = \text{pl_rade} / (\text{st_rad} \times 109.2)$ (1 Solar radius ≈ 109.2 Earth radii). After correction, the median radius ratio is 0.026, which is physically reasonable.

Missing value treatment: Following the “no-imputation principle” of Tang (2026g), missing values are retained as NaN, with no interpolation or filling. The missing rate for candidate executing factors entering exhaustive testing ranged from 0% (`radius_ratio`) to 27.8% (`st_dens`). Kolmogorov-Smirnov tests indicated distributional differences between retained and discarded samples ($p < 0.05$), which are candidly discussed as a limitation in §4.8. The coefficients of the optimal model on the complete sample and the full sample are exactly identical (all four coefficients match to six decimal places), demonstrating that the pattern of missingness does not affect coefficient estimation.

Final sample: 15,966 transiting planets, of which **15,540** possess complete data for all factors in the optimal model.

2.3 Candidate Factor Pool

Candidate executing factors (12): `radius_ratio` (planet-to-star radius ratio), `st_teff` (stellar effective temperature), `st_logg` (surface gravity), `st_mass` (stellar mass), `st_met` (metallicity), `pl_orbper` (orbital

period), `pl_orbsmax` (semi-major axis), `pl_orbincl` (orbital inclination), `pl_eqt` (equilibrium temperature), `pl_insol` (insolation flux), `sy_dist` (stellar distance), `st_dens` (stellar density). All are objective physical quantities produced by automated NASA pipelines, with no subjective scoring.

Candidate regime factors (3): discovery facility (`facility_group`: Kepler/TESS/K2/Ground), stellar spectral type (`spectral_class`: M/K/G/F/A), era group (`era_group`: Pre-Kepler/Kepler-Prime/K2/TESS).

2.4 Methodology: Four-Test Method + 5W2H + PDCA

This study strictly follows the Four-Test Method proposed by Tang (2026g), with the 5W2H and PDCA frameworks ensuring completeness and process closure.

First Test: Exhaustive Testing (Plan-Do). All 1- through 4-factor combinations of candidate executing factors are traversed and ranked by adjusted R^2 under the hard constraint of $VIF < 5$. The optimal combination wins automatically, without relying on subjective researcher selection. The test is repeated under the relaxed criterion of $VIF < 10$ to confirm robustness.

Second Test: Interaction Effect Testing (Core Evidence, Do—Modulation Dimension). Test whether the candidate regime factor modulates the exposure coefficient of executing factors on transit depth. A significant interaction effect is a necessary condition for identifying a regime factor.

Third Test: Chow Test (Core Evidence, Do—Structural Dimension). Using the candidate regime factor as the grouping variable, test whether the entire transit-depth equation exhibits a structural break between two sub-samples. Interaction effects and Chow tests corroborate each other; neither is sufficient alone.

Fourth Test: Auxiliary Evidence (Check). Includes permutation tests (dependent variable randomly shuffled 500 times), quantile regressions (testing whether effects are driven by extreme values), pseudo-out-of-sample prediction (Kepler training → TESS testing), and placebo tests (randomly shuffled facility labels). Granger causality testing is not applicable to cross-sectional data and is replaced by independent derivation from physical mechanisms.

5W2H Completeness Standard. Ensures the research question is completely answered across seven dimensions: What (what phenomenon was discovered), Why (why does it occur—physical mechanism), Who (who is responsible—identifying proxy variables), Where (within what scope does it hold), When

(where are the critical thresholds for regime switching), How (what is the transmission pathway), and How Much (what is the magnitude of the effect).

PDCA Process-Closure Standard. Ensures the research process completes the full cycle: Plan (design the testing scheme), Do (execute all tests), Check (examine alternative explanations and logical gaps), Act (convert conclusions into actionable criteria).

The dependent variable is `pl_trandep` (transit depth). All regressions employ HC3 heteroskedasticity-robust standard errors. Methodological details are provided in §2; all analyses are based on public data, and any researcher can independently reproduce the results.

3. Results

3.1 Exhaustive Testing: Optimal Combination of Executing Factors (Plan-Do)

Twelve candidate executing factors, $VIF < 5$ constraint, minimum sample size 500. The exhaustive procedure traversed 793 combinations. Results are shown in Table 1.

Table 1: Top 5 Models from Exhaustive Testing ($VIF < 5$)

Rank	Factor Combination	Sample Size	Adjusted R^2	Max VIF
1	<code>radius_ratio + pl_orbper + sy_dist</code>	15,540	0.691	1.47
2	<code>radius_ratio + sy_dist</code>	15,558	0.690	1.40
3	<code>radius_ratio + pl_orbper</code>	15,948	0.684	1.09
4	<code>radius_ratio</code>	15,966	0.683	—
5	<code>radius_ratio + st_teff + pl_orbper</code>	15,607	0.679	2.06

The optimal model `radius_ratio + pl_orbper + sy_dist` has the highest adjusted R^2 (0.691) among all $VIF < 5$ combinations, with a maximum VIF of only 1.47, well below the health threshold. Under the relaxed $VIF < 10$ criterion, this combination remains first, ruling out threshold selection effects on the conclusion.

The physical meaning of the optimal model is clear: `radius_ratio` is the geometric determinant of transit depth (transit depth is proportional to the square of the radius ratio), `pl_orbper` corrects for transit duration and orbital phase, and `sy_dist` corrects for photometric signal-to-noise.

Key observation: As a single factor, radius_ratio yields an adjusted R^2 of 0.683. If transit depth were a pure geometric measurement, this value should approach 1.0. The remaining 31.7% residual space is precisely where regime coupling may reside. The core task of this paper is to test whether this 31.7% residual is systematically modulated.

3.2 Interaction Effect Testing: The Discovery Facility Modulates the Exposure

Coefficient of radius_ratio (Do—Modulation Dimension)

Using Kepler as the baseline, we tested the interaction effects of TESS (fac_TESS), K2 (fac_K2), and ground-based telescopes (fac_Ground) with each executing factor in the optimal model.

Table 2: Facility Interaction Effect Test Results

Executing Factor	Regime Dummy	p-value	ΔR^2	Significance
radius_ratio	fac_Ground	6.28×10^{-20}	0.021	***
radius_ratio	fac_TESS	6.59×10^{-7}	0.048	***
sy_dist	fac_K2	1.06×10^{-5}	0.00017	***
pl_orbper	fac_K2	7.84×10^{-3}	0.00023	**
radius_ratio	fac_K2	9.91×10^{-3}	0.00080	**
sy_dist	fac_Ground	5.61×10^{-2}	0.00012	n.s.
pl_orbper	fac_TESS	1.80×10^{-1}	0.00040	n.s.
pl_orbper	fac_Ground	5.64×10^{-1}	0.00035	n.s.
sy_dist	fac_TESS	5.68×10^{-1}	0.00003	n.s.

Note: *** $p < 0.001$, ** $p < 0.01$, n.s. not significant. Baseline group is Kepler.

Five of the nine interaction terms are significant at the 5% level. Core finding: **The discovery facility systematically modulates the exposure coefficient of radius_ratio on transit depth.** The interaction ΔR^2 ranges from 0.00080 to 0.048. The ordering of modulation strength (TESS > Ground > K2) is consistent with the degree of physical difference in facility-star coupling.

For reference, Tang (2026g) found that the interaction incremental R^2 of the China-U.S. interest rate spread in finance was only 0.0016, yet was already argued to be a systematic signal. The modulation effects in astronomy are 13 to 30 times larger.

3.3 Chow Test: Structural Break in the Transit-Depth Equation Across Facilities

(Do—Structural Dimension)

Using the discovery facility as the grouping variable, we tested whether the optimal model's equation structure is consistent across pairs of facility groups.

Table 3: Chow Test Results Across Facilities

Comparison	Sample Sizes	F-statistic	p-value	Conclusion
Kepler vs TESS	14,637 vs 1,103	1,597.12	0.000	Structural break
Kepler vs Ground	14,637 vs 874	1,481.60	0.000	Structural break
Kepler vs K2	14,637 vs 680	201.54	0.000	Structural break

All three Chow tests yield p-values of zero. The ordering of F-statistics (TESS > Ground > K2) likewise follows the degree of physical difference. **The entire transit-depth equation exhibits an irreducible structural break across different facilities.**

3.4 5W2H Inquiry: Tracing the Physical Root (Check—Proxy Hypothesis Exclusion)

Is the discovery facility a true regime factor, or merely a proxy for a deeper physical variable? This is the core inquiry of the "Who" dimension in 5W2H. Kepler primarily observes G- and K-type main-sequence stars, while TESS contains more M- and F-type stars. The facility differences may simply reflect selection bias.

We replaced the discovery facility with stellar spectral type as the candidate regime factor and re-ran the interaction and Chow tests. Spectral types were extracted from the raw `st_spectype` field by taking the first letter (M/K/G/F/A), with G-type (Solar spectral type) as the baseline.

Table 4: Spectral Type Interaction Effect Test Results

Spectral Type	p-value	ΔR^2	Coefficient	Significance
spec_M	0.000000	0.156	8.27	***
spec_G	0.000000	0.066	6.43	***
spec_F	0.000000	0.040	6.63	***
spec_A	0.000425	0.0026	5.82	***
spec_K	0.003593	0.00068	4.73	**

Note: *** $p < 0.001$, ** $p < 0.01$. Baseline group is G-type.

All five spectral types exhibit significant interaction effects. The ordering of modulation strength ($M > G > F > A > K$) is fully consistent with stellar atmospheric physics: M-type stars have the lowest temperatures, the strongest molecular absorption bands, the most pronounced limb-darkening effects, and thus the strongest regime coupling.

Table 5: Chow Test Results Across Spectral Types

Comparison	Sample Sizes	F-statistic	p-value	Conclusion
G vs K	3,983 vs 6,550	315.21	0.000	Structural break
G vs F	3,983 vs 2,000	150.26	0.000	Structural break
K vs F	6,550 vs 2,000	104.25	0.000	Structural break
G vs M	3,983 vs 518	93.19	0.000	Structural break
K vs M	6,550 vs 518	63.41	0.000	Structural break
F vs M	2,000 vs 518	30.46	0.000	Structural break

All six pairwise comparisons yield Chow test p-values of 0.000000. The transit-depth equation exhibits structural breaks across different stellar spectral types as well.

The decisive test—after controlling for spectral type, the facility effect completely vanishes:

Table 6: Chow Test (Kepler vs TESS) After Controlling for Spectral Type

Spectral Type	Kepler Sample	TESS Sample	F-statistic	p-value	Conclusion
G-type	546	2,546	0.05	0.986	No difference
K-type	200	605	1.01	0.388	No difference
F-type	108	392	2.80	0.052	Marginally insignificant
M-type	19	98	Insufficient sample	—	—

Within all spectral types, the Chow test between Kepler and TESS is entirely non-significant. The previously observed “discovery facility as a regime factor” (Chow $F = 1,597$) is entirely driven by differences in the stellar spectral type distribution between facilities. Once spectral type is controlled for, the facility effect collapses from $F = 1,597$ to $F = 0.05$. **The discovery facility is not a regime factor—it is merely a proxy variable for the true regime factor, the stellar radiation field.** This is the first identification of a “proxy regime factor” within Factor Hierarchy Theory.

3.5 Locking in the Physical Root: Metallicity and Surface Gravity (Check—Physical Mechanism Confirmation)

If the stellar radiation field is the true regime factor, then the specific physical parameters that determine the structure of the radiation field—metallicity and surface gravity—should likewise exhibit the characteristics of regime factors. This addresses the “Why” and “How” dimensions of 5W2H.

3.5.1 Metallicity Amplification Effect

Stars were divided into three groups by metallicity [Fe/H]: metal-poor (< -0.2), Solar metallicity (-0.2 to $+0.2$), and metal-rich ($> +0.2$). Within each group, the spectral type interaction effect was tested.

Table 7: Metallicity Group × Spectral Type Interaction Effects

Metallicity Group	Sample Size	Spectral Type Interaction p-value	ΔR^2	Conclusion
Metal-poor	3,070	1.21×10^{-15}	0.0313	Extremely strong regime coupling
Solar metallicity	6,876	6.14×10^{-6}	0.0029	Moderate regime coupling
Metal-rich	1,169	0.428	0.0005	Regime coupling disappears

The spectral type interaction effect is extremely significant in the metal-poor group, moderate in the Solar metallicity group, and completely disappears in the metal-rich group. This inverse pattern is a precise mapping of stellar atmospheric physics: metal-poor stellar atmospheres consist almost entirely of hydrogen and helium, with a simple physical constitution, making the radiative transfer equation highly dependent on temperature and density structure (i.e., spectral type), yielding extremely strong regime coupling. Metal-rich stellar atmospheres are dominated by countless spectral lines of elements such as iron, magnesium, and silicon, causing the opacity across different wavelengths to become complex and homogenized, thereby “diluting” the influence of spectral type.

3.5.2 Surface Gravity Threshold Regression

Surface gravity ($\log g$) determines the pressure broadening in the stellar atmosphere, which in turn affects line shapes and the wavelength dependence of limb darkening. A grid search for the optimal threshold (between the 10th and 90th percentiles) yielded **$\log g = 4.64$** .

Table 8: Surface Gravity Threshold Regression Results

Threshold Value	Threshold Regression F	Threshold Interaction p-value	Conclusion
$\log g = 4.64$	14,645.36	4.63×10^{-9}	Regime switch

The threshold value of 4.64 falls precisely at the boundary between main-sequence stars ($\log g > 4.6$) and subgiants/giants ($\log g < 4.6$). **The radiation field properties undergo a qualitative change at $\log g = 4.64$.** The same planet, orbiting a main-sequence star versus a subgiant, would yield physically non-equivalent measured radii even if the temperatures were identical.

3.6 Temperature Threshold: The 4762 K Regime-Switching Critical Point (Check—Threshold Confirmation)

Using stellar effective temperature as a continuous variable, we performed a grid search for the regime-switching critical point. This addresses the “When” dimension of 5W2H.

Table 9: Stellar Effective Temperature Threshold Regression Results

Threshold Value	Threshold Regression F	Threshold Interaction p-value	Conclusion
4762 K	10,283.02	0.000	Regime switch

The threshold interaction coefficient is 2.88, $p = 0.000$. **Near 4762 K, the regime of the transit-depth equation switches.** This temperature falls precisely near the boundary between K- and G-type stars, marking the overall transition in molecular absorption bands, convection zone depth, and the wavelength dependence of limb darkening in stellar atmospheres. This is a complete methodological isomorphism with the China-U.S. interest rate spread ≈ 0 regime-switching threshold discovered by Tang (2026g) in finance.

3.7 Spectral Type × Metallicity Triple Interaction Effect (Check—Transmission Pathway Confirmation)

If metallicity is a modulator of regime coupling, it should amplify the modulation effect of spectral type on radius_ratio. This addresses the “How” dimension of 5W2H.

Table 10: Spectral Type × Metallicity × radius_ratio Triple Interaction Effect

Interaction Term	p-value	Conclusion
spec_Unknown × met_high × radius_ratio	1.82×10^{-7}	Metallicity amplifies spectral type modulation

The triple interaction effect is extremely significant. **Metallicity is not only a regime variable in itself; it also amplifies or attenuates the modulation effects of other regime factors.** The transmission pathway is now complete: stellar atmospheric chemical composition (metallicity) → changes in atmospheric opacity → altered wavelength dependence of limb darkening → different spatial distributions of the radiation field at different wavelength bands → detector response function selects specific bands → the measured “effective radius ratio” exhibits a regime-dependent systematic bias.

3.8 H-R Diagram Binning: The Spatial Distribution of Regime Coupling (Check—Spatial Distribution Confirmation)

Stars were binned two-dimensionally by temperature (Cool/Moderate/Hot) and surface gravity (main-sequence/evolved). Within each H-R diagram region, the spectral type interaction effect was tested. This addresses the “Where” dimension of 5W2H.

Table 11: Significant Spectral Type Interaction Effects Across H-R Diagram Regions

Temperature	Surface Gravity	Sample Size	Significant Interactions	Strongest ΔR^2
Moderate	High logg (MS)	5,972	1	0.0014
Moderate	Low logg (evolved)	3,802	1	0.0010
Hot	High logg (MS)	479	1	<0.0001

Regime coupling is concentrated in the “moderate temperature” region of the H-R diagram—precisely where F/G/K-type stars reside. This is the core region for both the Kepler and TESS planet samples.

3.9 Ultimate Evidence: Zero Cross-Facility Repeated Observations (Check—Sample Foundation Confirmation)

We examined whether any single planet in the NASA Exoplanet Archive had been observed by both Kepler and TESS.

Table 12: Cross-Facility Repeated Observation Statistics

Metric	Value
Total transiting planet records	18,116
Planets with multiple records	3,506
Planets observed by both Kepler and TESS	0

This “zero result” is not a data deficiency but an inevitable consequence of the physical design of the two survey missions: the Kepler prime mission fixed its pointing on the Cygnus-Lyra region (distant, faint stars), while TESS scans nearly the entire sky (nearby, bright stars). **The two facilities measure two physically non-overlapping populations of stars and planets.** The astronomical community possesses no empirical basis whatsoever for treating Kepler's “planetary radii” and TESS's “planetary radii” as directly comparable physical quantities.

3.10 Robustness Checks (Check—Statistical Artifact Exclusion)

Table 13: Comprehensive Robustness Check Results

Test Name	Core Metric	Result	Alternative Explanation Excluded
Permutation test	Empirical p-value	0.002	Statistical artifact
Quantile regression	0.25/0.50/0.75 quantiles	All p = 0	Extreme-value driving
Pseudo-out-of-sample prediction	RMSE amplification	4.4×	Cross-regime inter-predictability
Placebo test	Chow F collapse magnitude	98.3% reduction	Arbitrary grouping effect

Permutation test. The dependent variable (transit depth) was randomly shuffled 500 times, and the fac_Ground × radius_ratio interaction was re-tested each time. The median interaction p-value across the 500 random permutations was 0.5185, whereas the true p-value is 6.28×10^{-20} . The empirical p-value is 0.002, indicating that the probability of random noise producing an interaction effect of equal strength does not exceed 0.2%.

Quantile regression. Quantile regressions were performed at the 0.25, 0.50, and 0.75 quantiles of transit depth. The coefficients for fac_Ground × radius_ratio were 14.93, 15.04, and 13.24 respectively (all p = 0.000), and for fac_TESS × radius_ratio were 7.09, 7.38, and 6.74 (all p = 0.000). The interaction effects are highly stable across all quantiles and are not driven by extreme values.

Pseudo-out-of-sample prediction. The optimal model was fitted on Kepler data as the training set and tested on TESS data. The training-set RMSE was 0.178, while the test-set RMSE was 0.782—an amplification of 4.4 times. The mean prediction error was 0.167, systematically positive (the Kepler-trained model systematically underestimates transit depth on TESS data). The t-test for zero mean prediction error yields $p = 0.000$.

Placebo test. After randomly shuffling facility labels, the Chow test was re-run. The true-grouping F-statistic of 1,597 collapsed to 27.35 under random grouping (a 98.3% reduction), demonstrating that the genuine facility effect is far stronger than anything produced by arbitrary grouping.

4. Discussion

4.1 Paradigm Shift: Planetary Radius from “Intrinsic Quantity” to “Relational Quantity”

The core finding of this study is not the “discovery of a new planet” or the “correction of a model parameter,” but rather the redefinition of a foundational concept in exoplanet science:

Transit depth is not an intrinsic property of a planet, but a “relational quantity” arising from the coupling of the planet, the stellar radiation field, and the detector response.

This finding rests on the following irreducible chain of evidence:

(1) `radius_ratio` as a single factor explains only 68.3% of the variance. The remaining 31.7% residual is not random noise, but a structural signal systematically modulated by stellar spectral type, metallicity, and surface gravity.

(2) The discovery facility (Kepler/TESS) appears as a “regime factor,” but this effect completely vanishes after controlling for stellar spectral type (Chow F collapses from 1,597 to 0.05, $p = 0.99$). The facility is merely a **proxy variable** for the true regime factor, the stellar radiation field. This is the first identification of a “proxy regime factor” form within Factor Hierarchy Theory.

(3) Stellar spectral type, metallicity, and surface gravity were all tested and confirmed as significant regime factors. The physical root of regime coupling is locked in as stellar atmospheric opacity (the

metallicity effect), pressure broadening (the surface gravity effect), and overall radiation field structure (the spectral type effect).

(4) A temperature of 4762 K and $\log g = 4.64$ constitute a double critical point for regime switching, on either side of which the radiation field properties undergo a qualitative change.

(5) The Kepler and TESS planet samples are physically non-overlapping—not a single planet has had its radius measured by both facilities.

4.2 The Ultimate Diagnosis of TDPP

The essential paradox of TDPP is that instrumental precision is improving, yet the discrepancies between measurement results are increasing. The mainstream explanation in astronomy is insufficient LD models, and the proposed solution is to fix the models.

The diagnosis of this study is entirely different: **TDPP is not “error” but “structure.”** Astronomers are chasing a physically non-existent, cross-regime universal “planetary radius.” When measurement precision becomes sufficiently high, the signal of regime coupling emerges from the noise—this is the physical root of the “more precision, more discrepancy” paradox.

4.3 Implications for Exoplanet Statistics (Act—Actionable Criteria)

If the conclusions of this paper hold, the following inferences in exoplanet science need to be re-examined:

Planetary radius distribution functions. The radius distributions derived from Kepler and from TESS are based on two physically non-overlapping stellar samples and two different measurement regimes. They can neither be directly merged nor directly compared.

Planetary density classification. The boundary between super-Earths and mini-Neptunes lies at approximately 1.6 Earth radii. If TESS-measured radii differ systematically from Kepler-measured radii, the same planet could be classified into different categories by the two missions.

Habitable-zone planet radius precision. The JWST is measuring atmospheric compositions of habitable-zone planets. If the planet's radius itself is regime-dependent, the input parameters for atmospheric models are non-unique, and the derived atmospheric compositions may harbor systematic biases.

This paper provides the astronomical community with an actionable criterion: **before merging transit data from multiple facilities or multiple stellar types, one should conduct interaction effect and Chow tests using candidate regime factors as grouping variables. If both are significant, the existence of a regime switch should be acknowledged, and separate modeling should be adopted rather than forced uniform calibration.**

4.4 Methodological Contribution: Cross-Disciplinary Validation of Testability and the Discovery of the Proxy Regime Factor

This study is the first independent cross-disciplinary empirical test of the Principle of Testability proposed by Tang (2026g) beyond finance. The Four-Test Method replicated a factor hierarchy structure in astronomical data that is completely isomorphic to that found in finance:

Table 14: Comparison of Factor Hierarchy Structures in Finance and Astronomy

Dimension	Finance (Tang, 2026g)	Astronomy (This Paper)
Regime factor	China-U.S. interest rate spread	Stellar spectral type, metallicity, surface gravity
Executing factors	VIX, DXY, TIPS, etc.	radius_ratio, pl_orbper, sy_dist
Regime-switching critical point	Spread ≈ 0	Temperature ≈ 4762 K, $\log g \approx 4.64$
Interaction effect	spread $\times$ VIX significant	Spectral type $\times$ radius_ratio significant
Chow test	Positive/negative spread groups significant	Spectral type groups significant
Proxy regime factor	Not discovered	Facility (replaced by spectral type)

The astronomical results go a step beyond the financial ones: by demonstrating that the facility effect vanishes after controlling for spectral type, this paper has, for the first time within the framework of Factor Hierarchy Theory, identified a **proxy regime factor**. Why was no proxy regime factor discovered in finance? Because the China-U.S. interest rate spread is a direct product of central bank policy will—it is an **institutional variable** with no deeper physical variable manipulating it behind the scenes. In astronomy, by contrast, the facility is a human-chosen observational tool—it is an **instrumental variable**, and deeper physical roots (the stellar radiation field structure) naturally determine its manifestation. The proxy hierarchy of regime factors depends on whether the system is institution-driven or physically driven. This distinction enriches the structure of Factor Hierarchy Theory.

4.5 From Factor Hierarchy Theory to Factor Hierarchy Law

Tang (2026g) proposed Factor Hierarchy Theory across five major financial markets and candidly stated in that paper that “cross-disciplinary independent verification awaits subsequent research.” The empirical results of this paper provide the first independent verification of Factor Hierarchy Theory in astronomy.

Given that the theory has now independently replicated the same hierarchical structure across five financial markets and exoplanetary transit measurements—two completely unrelated disciplines, the same exhaustive search → interaction → Chow testing procedure, the same regime-executing hierarchy, and the same regime-switching critical points—this paper proposes elevating this theory to the status of **Factor Hierarchy Law**.

4.6 The Relationship Between 5W2H, PDCA, and the Four-Test Method

The empirical process of this paper also constitutes the first complete implementation of the 5W2H and PDCA frameworks as completeness companions to the Four-Test Method. Table 15 summarizes the functional division of labor among the three.

Table 15: Functional Division of the Four-Test Method, 5W2H, and PDCA

Framework	Function	Implementation in This Paper
Four-Test Method	Testing procedures	Exhaustive search → Interaction → Chow → Auxiliary evidence
5W2H	Question completeness	What (regime coupling exists), Why (stellar atmospheric physics differences), Who (spectral type/metallicity/surface gravity are regime factors; facility is a proxy), Where (moderate-temperature region of H-R diagram), When (4762 K and log g = 4.64 double critical points), How (metallicity → opacity → limb darkening → detector selection transmission chain), How Much (ΔR^2 from 0.00068 to 0.156)
PDCA	Process closure	Plan (designed 12-factor exhaustive scheme), Do (exhaustive → interaction → Chow → spectral type → metallicity → surface gravity → H-R diagram → triple interaction → zero-overlap), Check (permutation test, quantile regression, controlling for spectral type, placebo test), Act (proposed pre-merger transit data inspection criterion)

This provides the first cross-disciplinary empirical case study for the “completeness companion to testing procedures” proposed by Tang (2026g).

4.7 Implications for Stellar Physics

If the radiation field structures of stars of different spectral types, metallicities, and surface gravities undergo a structural break on either side of critical points (Chow $F = 14,645$ and $10,283$), then this issue is not merely a technical question of "how to measure planetary radius," but a physical question of "whether the stellar radiation field itself possesses regime dependence." Current stellar atmosphere models (ATLAS, MARCS, PHOENIX, etc.) are all built upon underlying assumptions of one-dimensional static atmospheres, local thermodynamic equilibrium, and mixing-length theory. These assumptions are good approximations for Solar-type stars but may systematically fail for M dwarfs (dominated by molecular opacity) and giants (where non-LTE effects are significant). We recommend that subsequent studies use Gaia's independent photometric and parallax data to cross-validate stellar parameters derived from transit fitting, employing the Four-Test Method to detect whether systematic regime-dependent deviations exist between model outputs and independent measurements.

4.8 Limitations and Candid Disclosures

Sample missingness bias. KS tests indicate systematic distributional differences between retained and discarded samples ($p < 0.05$). This is a natural consequence of the differing survey strategies of Kepler and TESS. This paper has demonstrated, via exact coefficient identity between the optimal model on the complete sample and the full sample (all four coefficients match to six decimal places), that the missingness pattern does not affect coefficient estimation.

Radius ratio measurement error. The raw `pl_rade` median relative error is 23.5%, and `st_rad` is 20.6%. However, `radius_ratio` ranked first in exhaustive testing (single-factor $R^2 = 0.683$), indicating that its physical signal overwhelms the measurement noise.

Spectral type classification granularity. This study used the first letter of the spectral type as the classification criterion, without subdividing into subtypes or luminosity classes. Finer classification may require consulting raw spectral data and is left for future research.

Cross-researcher replication. The methodological details of this study are provided in §2; all analyses are based on NASA public data. Independent replication of the conclusions by other researchers awaits verification.

5. Conclusion

This paper has applied the Four-Test Method, 5W2H, and PDCA frameworks proposed by Tang (2026g) to conduct a presupposition-free analysis of public transit data from the NASA Exoplanet Archive, yielding the following conclusions:

(1) **Exhaustive testing:** radius_ratio + pl_orbper + sy_dist is the optimal combination of executing factors for transit depth (adjusted $R^2 = 0.691$, $VIF < 5$), with results consistent under dual VIF thresholds.

(2) **Discovery of regime coupling:** Transit depth is not an intrinsic property of a planet, but a “relational quantity” arising from the coupling of the planet, the stellar radiation field, and the detector response. radius_ratio as a single factor explains only 68.3% of the variance; the remaining 31.7% residual is systematically modulated.

(3) **Discovery of the proxy regime factor:** The discovery facility (Kepler/TESS) is a proxy regime factor for the stellar radiation field. After controlling for spectral type, the inter-facility Chow F collapses from 1,597 to 0.05 ($p = 0.99$). This is the first identification of a proxy regime factor form within Factor Hierarchy Theory. The proxy hierarchy of regime factors depends on whether the system is institution-driven or physically driven.

(4) **Locking in the physical root:** Stellar spectral type, metallicity, and surface gravity are the true regime factors. The physical root of regime coupling is stellar atmospheric opacity (the metallicity effect), pressure broadening (the surface gravity effect), and overall radiation field structure (the spectral type effect).

(5) **Double critical point for regime switching:** A temperature of 4762 K ($F = 10,283$) and surface gravity $\log g = 4.64$ ($F = 14,645$) constitute the physical thresholds for regime switching, corresponding to the overall transition in molecular absorption bands and convection zone depth in stellar atmospheres.

(6) **Cross-facility sample physical non-overlap:** The Kepler and TESS planet samples have zero overlap. The astronomical community possesses no empirical basis for treating their “planetary radii” as directly comparable physical quantities.

(7) **Cross-disciplinary methodological validation and theoretical elevation:** The Four-Test Method + 5W2H + PDCA have independently replicated the same factor hierarchy structure in finance and astronomy. Given that the theory has now received independent verification across two completely unrelated disciplines, this paper proposes elevating Factor Hierarchy Theory to the status of **Factor Hierarchy Law**.

Core Conclusions: (1) Transit depth is a relational quantity arising from the coupling of planet, stellar radiation field, and detector response, not an intrinsic property of the planet. (2) The discovery facility is a proxy regime factor for the stellar radiation field; the true regime factors are stellar spectral type, metallicity, and surface gravity. (3) The concept of “planetary radius” as used in astronomy for four decades lacks physical universality in cross-regime comparisons.

The ultimate root of the Transit-Depth Precision Problem is not that measurements are insufficiently precise, but that “precise measurement” itself has a physical limit in the face of regime coupling. This paper provides the astronomical community with a procedural criterion: before merging multi-facility transit data, one should conduct interaction effect and Chow tests using candidate regime factors as grouping variables; if both are significant, the regime switch should be acknowledged, and separate modeling should be adopted rather than forced uniform calibration. This criterion and its methodological foundation—the Four-Test Method + 5W2H + PDCA—have been independently verified in finance and astronomy, demonstrating their independence from the physical assumptions of any specific discipline.

Data Availability Statement

All data in this study were obtained from the NASA Exoplanet Archive (<https://exoplanetarchive.ipac.caltech.edu>), downloaded via the TAP synchronous query interface. The data are fully public and require no application. The raw file contains 39,913 rows and 355 columns. No pre-screening was applied to the raw file. Methodological details are provided in §2; all analyses are based on public data, and any researcher can independently reproduce the results.

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